The Functional Mock-up Interface Beginners’ Tutorial

Christian Bertsch, Bosch Research, FMI Project Leader

Online Materials:

Further Contributors:
Cinzia Bernardeschi, University of Pisa
Claudio Gomes, Aarhus University
Maurizio Palmieri, University of Pisa
Torsten Sommer, Dassault Systèmes

https://github.com/modelica/fmi-beginners-tutorial-2026/tree/main

Agenda

Part 1: Introduction to FMI (40Min)

- Motivation / History
- How does FMI work?
- Tool support
- New features in FMI 3.0

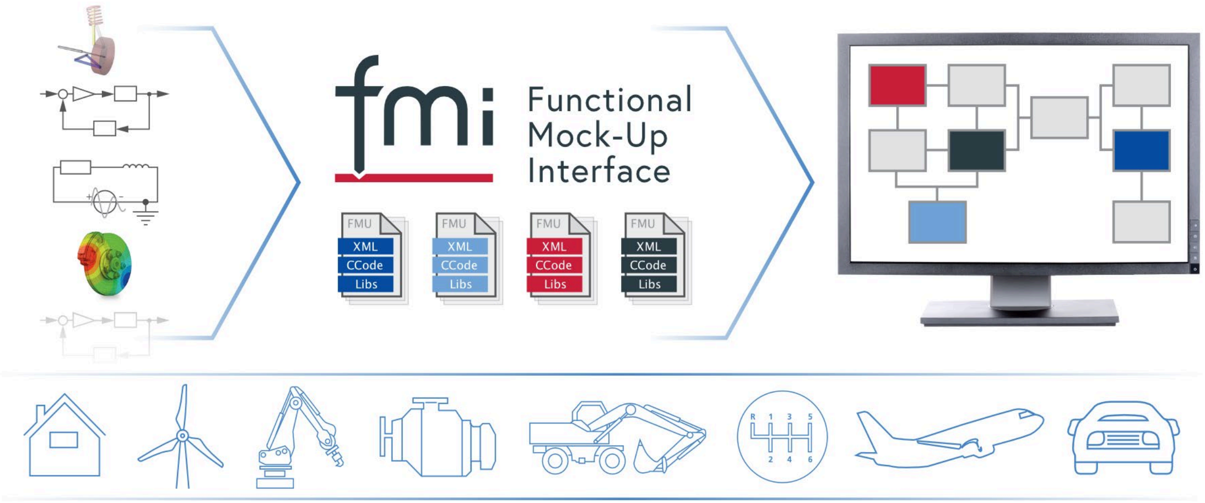
Part 2: Working with (single) FMUs (45 min)

Part 3: Interacting with multiple FMUs (45 min)

Part 4: Closing Session (10 min)

Q&A

Introduction to FMI



Online Materials:

https://github.com/modelica/fmi-beginners-tutorial-2024/tree/main

Motivation, Timeline

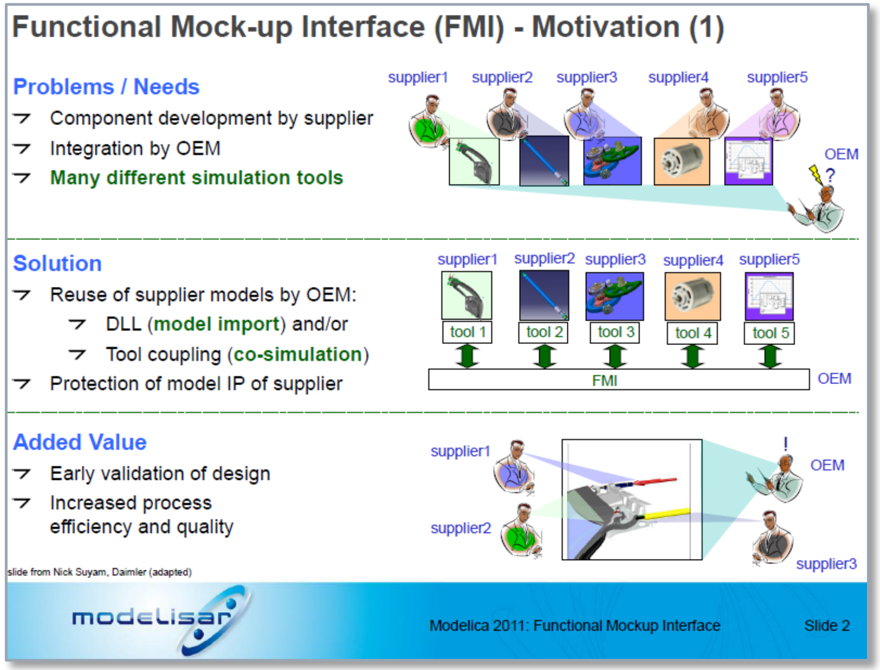
Motivation

- Define **a tool-independent**, **free-to-use** standard for exchange and co-simulation of models between different simulation tools
- **Provide models in a containerized form that** allow for the **deployment to different targets**
- **Decouple Know-How**
  - between producers and users of FMUs
  - between different specialized engineers and software programmers
- Massive **Re-use** of modelling investment
- **IP Protection** possible vial black-box model exchange

Timeline : Modelisar Project → Modelica Association Project FMI

- FMI 1.0 (and most part of FMI 2.0) was developed in the publicly funded project Modelisar
- 2008-2011: MODELISAR project
- 2011: Release of FMI 1.0
- 2012: Foundation of MAP FMI in MA

  Members: see [FMI Webpage]
- 2013: Release of FMI 2.0 → *focus of this tutorial*
- 2022: Release of FMI 3.0
- 2024, 25, 26: Release of FMI Layered Standards (FMI-LS-XCP, FMI-LS-BUS, …)



Versioning of FMI

FMI uses semantic versioning:

Major releases

- not backwards or forward compatible changes

e.g. **„FMI 3.0.2“**

Minor releases

- Can introduce new features
- Backwards compatible
  „FMI 3.0 FMUs will be valid FMI 3.1 FMUs“

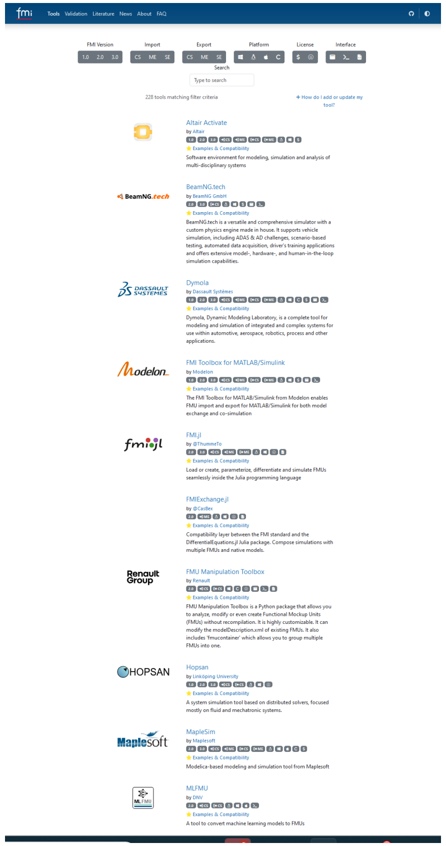
Bugfix/Maintenance releases

- No new features, only bugfixes and clarifications possible
- Backwards and forward compatibility

The Functional Mock-Up Interface today

The Functional Mock-up Interface is the leading free, vendor-neutral standard for the exchange and co-simulation of simulation models

- Current releases: **FMI 2.0.5** and **FMI 3.0.2**
- **280+ tools** and libraries support FMI (https://fmi-standard.org/tools/)



The Modelica Association

The Modelica Association is a non-profit, non-governmental organization with the aim of **developing and promoting open and coordinated standards for system modelling and simulation.**

MA Projects:

- A project to develop, standardize, and promote Modelica, a language to model and simulate multi-domain cyber physical systems in a convenient way. Modelica Language
- A project to develop, maintain, and promote the open source Modelica libraries and model components in many domains based on standardized interface definitions. Modelica Libraries
- A free standard that defines a container and interface to exchange dynamic simulation models using a combination of XML files, binaries, and C code. fmi-standard.org
- A tool independent standard to define complete systems of one or more FMUs including its parameterization that can be transferred between simulation tools. ssp-standard.org
- A platform and communication medium independent standard to integrate models and real-time systems into simulation environments. dcp-standard.org
- A standard to seamlessly integrate physics design models of systems with their electronic controls development. efmi-standard.org

The Modelica Association FMI Project

The development of the FMI Standard is organised as the Modelica Association Project *Functional Mock-up Interface* under the roof of the Modelica Association. It follows the Project Rules approved by the FMI Steering Committee.

Project Leader and Deputy:
Christian Bertsch (BOSCH) and Torsten Sommer (Dassault Systèmes)

Members of the Steering Committee:
AVL List, BOSCH, Dassault Systemes, dSPACE, ESI Group, Maplesoft, Modelon, PMSF, Siemens PLM, Synopsys

Further Contributing Members:
Aarhus University, ABB, Altair, Akkodis, AMEPERE, Ansys, Augsburg University, Beckhoff, Boeing, Danfoss, DLR, EKS INTEC, ETAS, Fraunhofer IEM, IAV, ITK Engineering, iVH, JuliaComputing, LTX, MachineWare, Renault, Saab Group, Virtual Vehicle Research, Wolfram MathCore AB, TLK Thermo, tracetronic, TU Dresden *and all Steering Committee Members*

Members of the Advisory Committee: AIRBUS, blue automation, Claytex, COMSOL, DNV, Fraunhofer (IIS/EAS First, SCAI), GM Motorsports, KEB Automation, LBL, NVIDIA, Knorr-Bremse Rail Vehicle Systems, MathWorks, Open Modelica Consortium, Samares Engineering, SINTEF Nordvest, University of Halle, Volkswagen, Volvo Autonomous Solutions, VTI, *and all Contributing and Steering Committee Members*

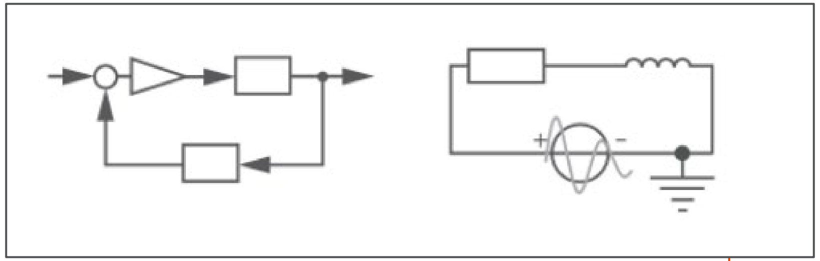
within the Modelica Association

FMI: Technical Fundamentals

“How does FMI work?”

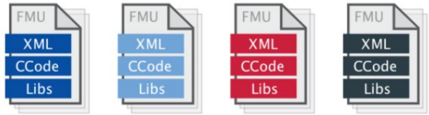
Exporting vs. Importing tool

Exporting tools



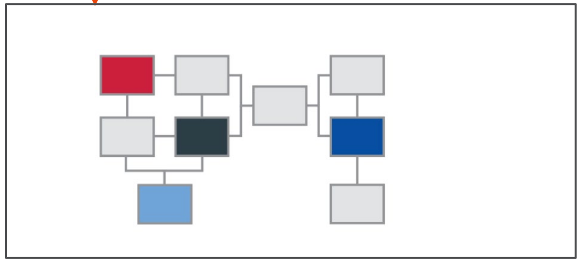
1. develop model
2. export as **Functional Mockup Unit (FMU)** accordingg to the **FMI Standard**

FMU



3. import FMU

4. (co-) **simulate model**

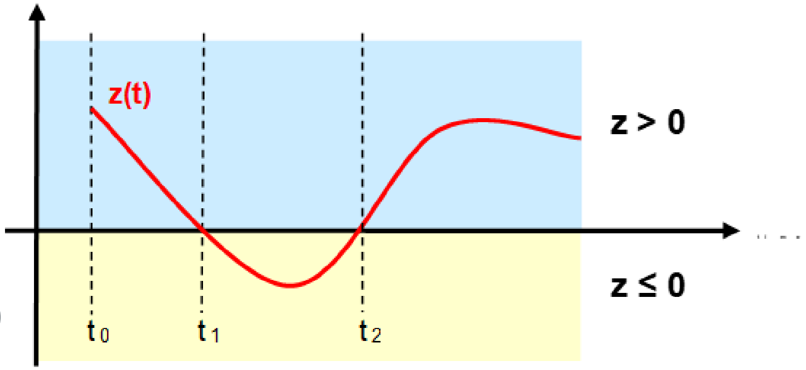


Importing tool

Models in Scope

Original Scope of FMI:

- Ordinary differential equations (**ODEs**) with **events**
  - You need a **numerical solver** for their solution
- Distinction between continuous and discrete variables
- There is a notion of **time** (or more general „independent variabl2)



Extended scope:

- Purely algebraic equations
- Complex discrete behavior with clocks and model partitions (FMI 3.0)
- A very broad scope of use cases such as
  - virtual electronic control units (ECUs),
  - AI models
  - ...

Co-Simulation (CS)

Co-Simulation: (from the FMI 3.0.2 glossary)

- **Coupling of several *simulation programs*** in order to compute the global behavior of a system that consists of several subsystems.
- Subsystems are coupled in the sense that the **behavior of each subsystem depends on the behavior of the remaining subsystems**, so that the co-simulation must be **computed in a step-by-step fashion**.
- Each simulation program is responsible for computing the behavior of a subsystem, using the outputs produced by the other simulation programs.

- There can be an **additional error** introduced by the co-simulation, that has to be limited to an acceptable size by using a **suitable co-simulation algorithms and communication pattern** (e.g. step size)

The FMI Standard

The FMI Standard defines the “Functional Mock-up Interface” FMI, an interface between a model and a simulation environment.

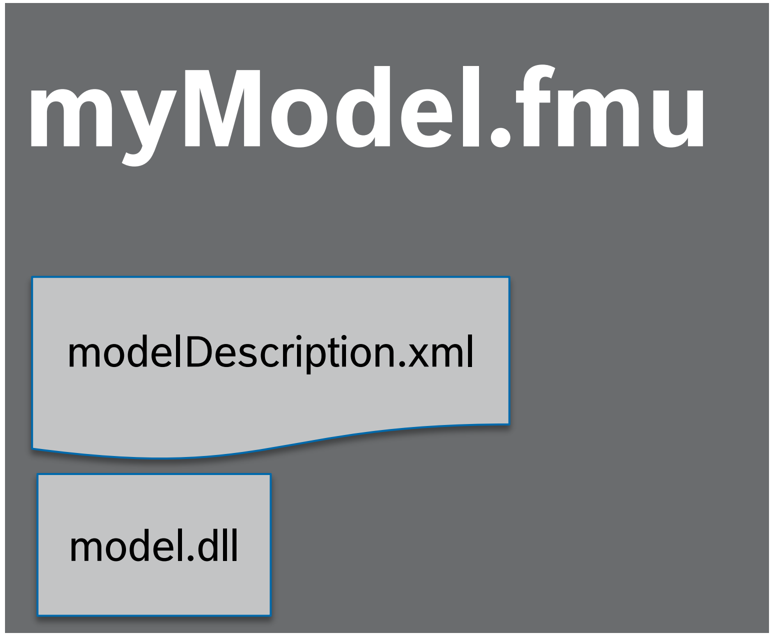
It consists of the definition of

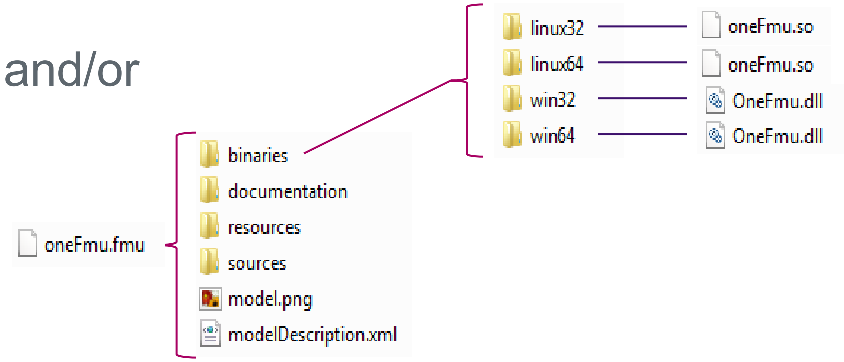
- a **C-API** (application programming interface)
- an **interface description** (modelDescription.xml) according to a defined schema
- the **definition of an exchangeable unit**, an **“FMU”** (Functional Mock-up Unit), technically a zip file

The FMI Standard only defines the interface of a single FMU,
not the co-simulation algorithm or solver for multiple FMUs!

FMU – „Functional Mock-up Unit“

- A model container, that can be distributed
- Technically it is realized a zip File, with ending „.fmu“
- Content:
  - **modelDescripotion.xml:**
    meta-data with information on the model variables, interface, capabilities and to a limited extent model structure
  - **Model representation**
    - **Binaries** for one or multiple platforms and/or (“black box”), and/or
    - **Source code** (e.g C-Code)
  - Optionally: Resources, documentation, Icons, port definitions
  - /extra information (defined in layered standards)

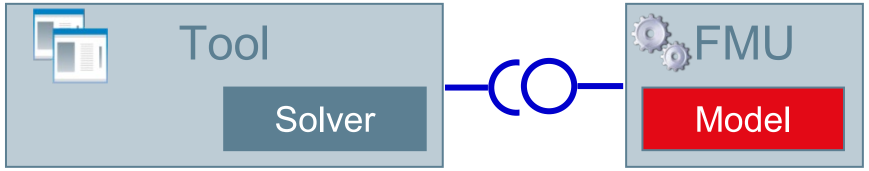




FMU for Model Exchange vs. Co-Simulation (FMI 2.0)

Model Exchange (ME)

- Importing tool provides the solver.



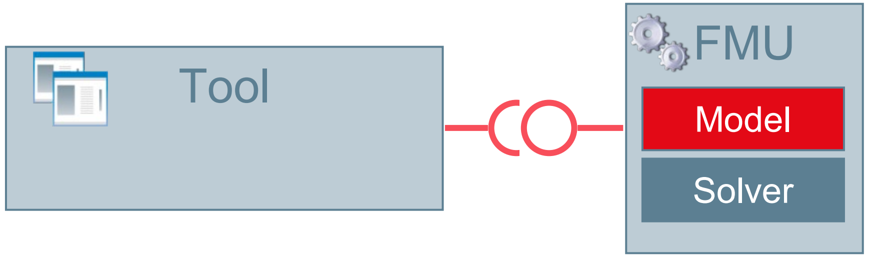
Properties:

- Very tight integration of model in simulation tool
- Complex interface between importing tool and model
- Importing tool must provide a suitable solver for the model

- Used e.g. for inclusion of Modelica support in non-Modelica tools (using the solver of the importing tool)

Co-Simulation (CS)

- Exporting tool provides the solver.



Properties:

- Tight coupling of model and a suitable solver
- Simpler interface between importing tool and model
- Freedom in the selection of a co-simulation algorithm and communication timestep to reach a stable and accurate solution

- Used in most industrial applications

Co-Simulation (CS) Interface

- Communication timestep can be different from internal steps (e.g. Variable step solver)
- Calling sequence CS
  - Set inputs: **`fmi2SetXXX(...)`**
  - Trigger calculation until next communication point: **`fmi2doStep(...)`**
  - Get outputs: **`fmi2GetXXX(...)`**

For an implementation in C, see e.g. fmusim in the Reference FMUs:
https://github.com/modelica/Reference-FMUs/blob/main/fmusim/fmusim_fmi2_cs.c

You can view the calling sequence `with  fmusim --log-fmi-calls`

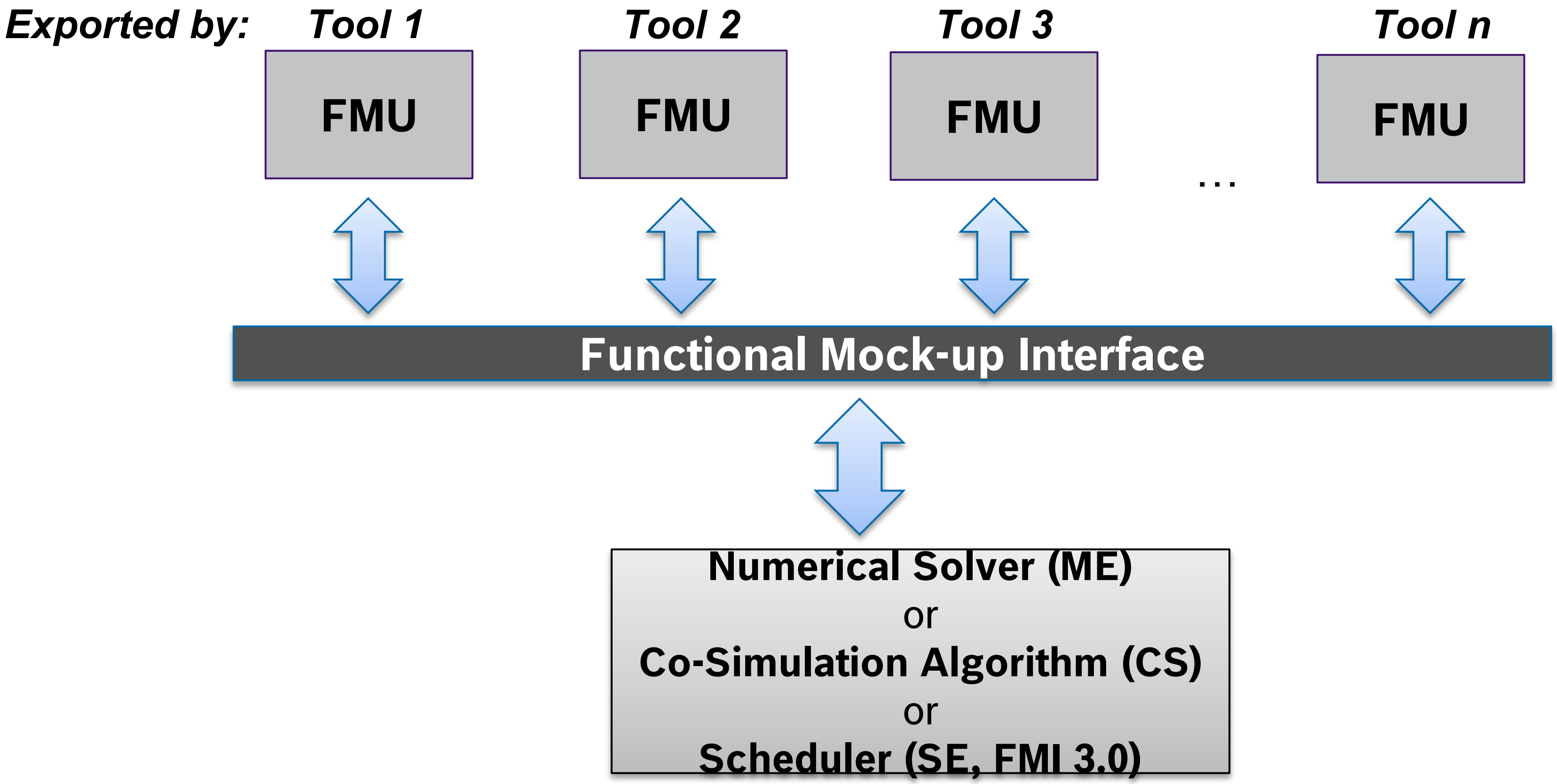
- The co-simulation algorithm is not part of the FMI standard. It is responsible for:
  - advancing the overall simulation time,
  - exchange input and output data,
  - triggering of input clocks, and handling events.
- Internally the FMU can have different timesteps (e.g. a variable step solver)

Model Exchange (ME) Interface

Main idea: the ME interface exposes the **right hand side** (RHS) of a **hybrid ODE** to the external solver.

- The **importer controls the data exchange** and the synchronization between FMUs.
- **The solver algorithm itself is not part of the FMI standard**. It is responsible for:
  - advancing the overall simulation time,
  - exchange input and output data,
  - computation of continuous state variables by time integration,
  - triggering of input clocks, and
  - handling events.

Simulation of multiple FMUs



Not defined by FMI Standard, but by the importing tool

The modelDescription.xml

It contains all static information about the FMU

- Definition of **supported interface kinds** (ME, CS, SE)
- **Model variables**
  - Inputs, outputs, parameters
- Certain information on the **model structure**

```xml
<?xml version="1.0" encoding="UTF-8"?>
<fmiModelDescription
  fmiVersion="2.0"
  modelName="BouncingBall"
  description="This model calculates the trajectory, over time,
  generationTool="Reference FMUs (development build)"
  guid="{1AE5E10D-9521-4DE3-80B9-D0EAAA7D5AF1}"
  numberOfEventIndicators="1">

  <ModelExchange
    modelIdentifier="BouncingBall"
    canNotUseMemoryManagementFunctions="true"
    canGetAndSetFMUstate="true"
    canSerializeFMUstate="true">
    <SourceFiles>
      <File name="all.c"/>
    </SourceFiles>
  </ModelExchange>
```

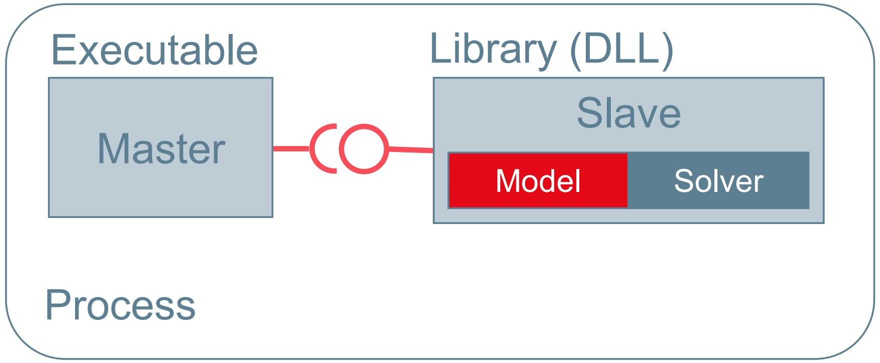
- Example: https://github.com/modelica/Reference-FMUs/blob/main/BouncingBall/FMI2.xml

- Attributes and capability flags, such as
  - “needsExecutionTool”
  - …

FMI for Co-simulation Tool Wrapper Variant

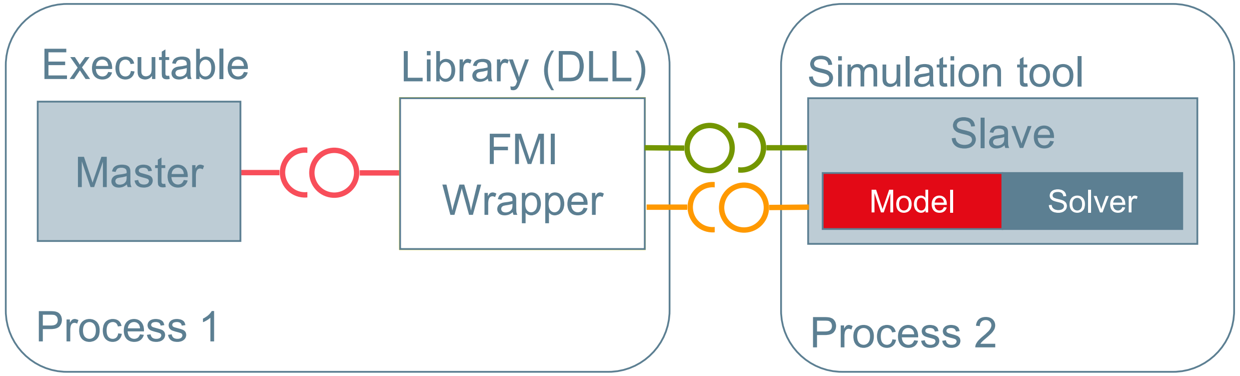
Co-simulation stand-alone FMU

- self-contained FMU



Co-simulation FMI Tool Wrapper

- FMU dependent on an existing tool installation



Attribute: “needsExecutionTool”

FMI Tool Support

https://fmi-standard.org/tools/: FMI supported by >= 250 Tools

- Exporting tools
  - Modeling and simulation tools
  - Model-based software creation
  - Wrapping of algorithms, virtual ECUs
  - …

- Importing tools
  - Modeling and simulation tools
  - Co-simulation tools and integration platforms
  - Implementations for programming and scripting languages
  - ….

Compatibility Information and validation tols

- Replaces the cross-check that helped to improve he maturity of FMI 1.0 and 2.0 supporting tools.
- Information on on how FMI export and import have been tested provided by the tool vendors and linked in the tools list
- Tools providing compatibility information are marked with a golden star and listed on top of the tools list https://fmi-standard.org/tools/

- Four FMU checkers provided on https://fmi-standard.org/validation/
  - FMU Check (static check, web-based, based on fmpy)
  - fmpy
  - fmusim from the Reference FMUs
  - FMI-VDM-Model Checker

Licenses and licensing mechanism

- The **FMI standard deals with the technical part** of model-exchange and co-simulation only.

- There might be other **legal or technical restrictions**
  - Exporting tools might use a **licensing mechanism** such as FlexNet
  - **Other legal obligations**, e.g., w.r.t. to the distribution of FMUs might be imposed by the exporting tools.

- **License information** can be stored in the documentation folder

New features in FMI 3.0

Motivation for FMI 3.0 (Modelica Conference 2021)

180+ tools support FMI now: many users now, many new use-case requests:

- Virtual Electronic Control Units (vECUs):
  - FMI 2.0 works well for physics simulations: better support for vECUs is needed
- Advanced Co-Simulation
  - Co-Simulation is the more popular interface type: improved co-simulation methods are needed to improve performance and accuracy
- Multi-FMU simulations are getting more common
  - Events are necessary in complex control systems
  - Events must be synchronized across FMUs
- New ML and AI applications
  - More derivatives computations is required

vECU

Co-Sim

Events

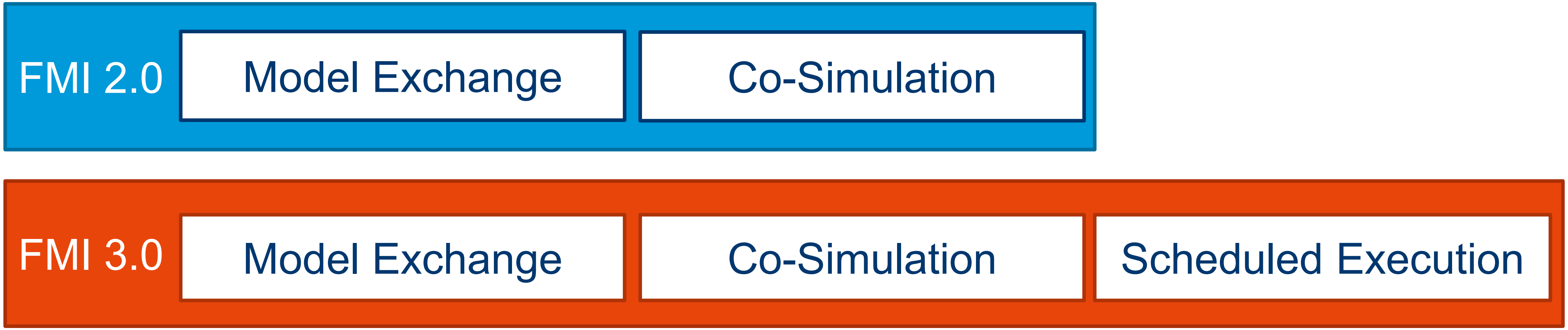
AI

API Efficiencies

FMI 3.0: New Interface Type – Scheduled Execution

vECU

Events



Scheduled Execution allows coupling several FMUs with one, external scheduler (OS)

FMI 3.0: Main Improvements

- Event mode for Co-Simulation
- Intermediate variable update
- Clocks
- New variable types
- Array variables
- Terminals and icon
- FMI for Scheduled Execution (SE)
- Preparation for layered Standards

Performance

Accuracy

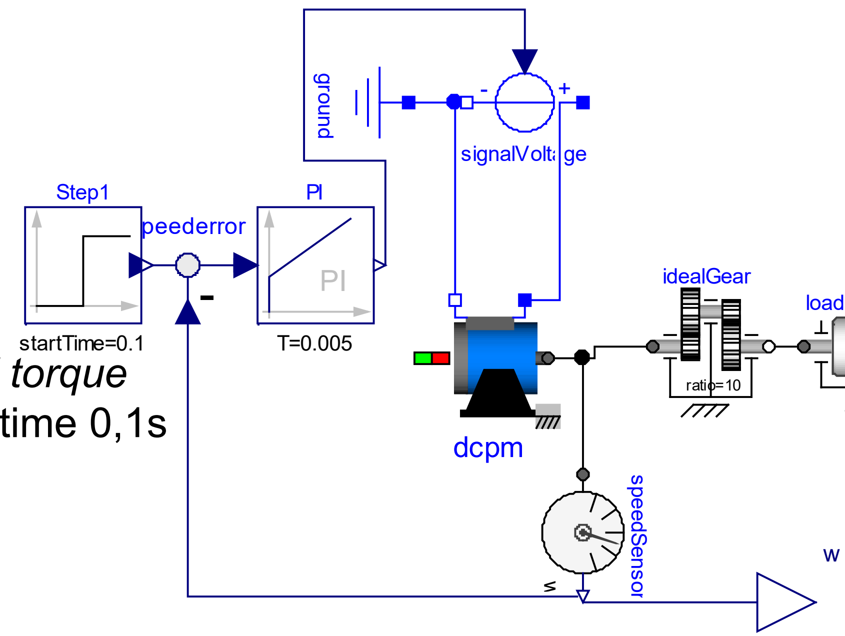
New Application

… more in part 4!

Demo:
Example model for hands-on part

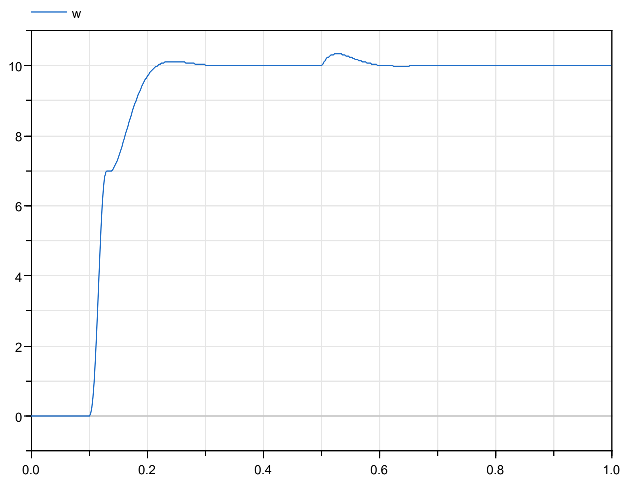
Example model: Controlled Motor Drive

Motor Drive with PI control modelled in Modelica



Desired torque
Step at time 0,1s

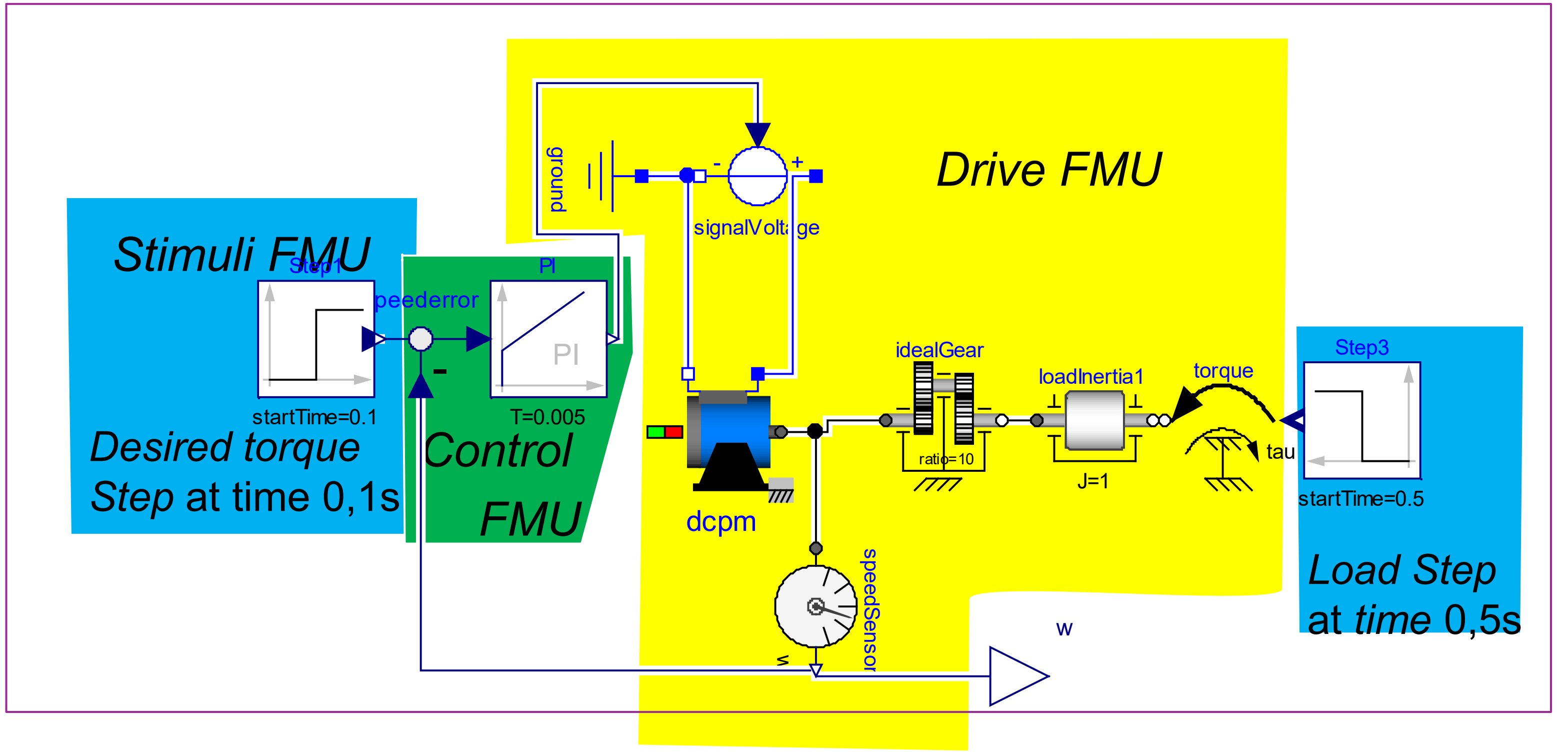
Load Step
at *time* 0,5s



Reference solution
Angular speed

Example model: Controlled Motor Drive

Model exported as one overall FMU or split as three FMUs.

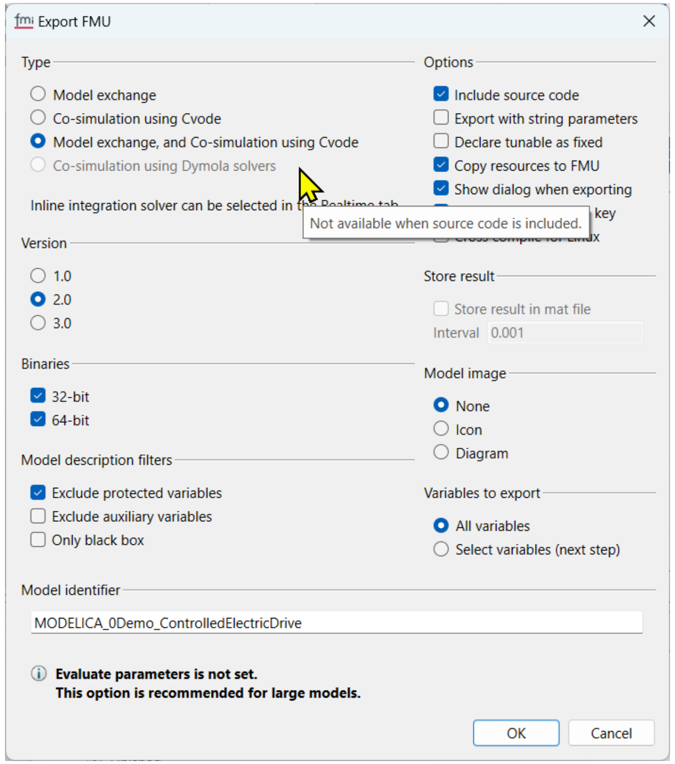


Overall FMU

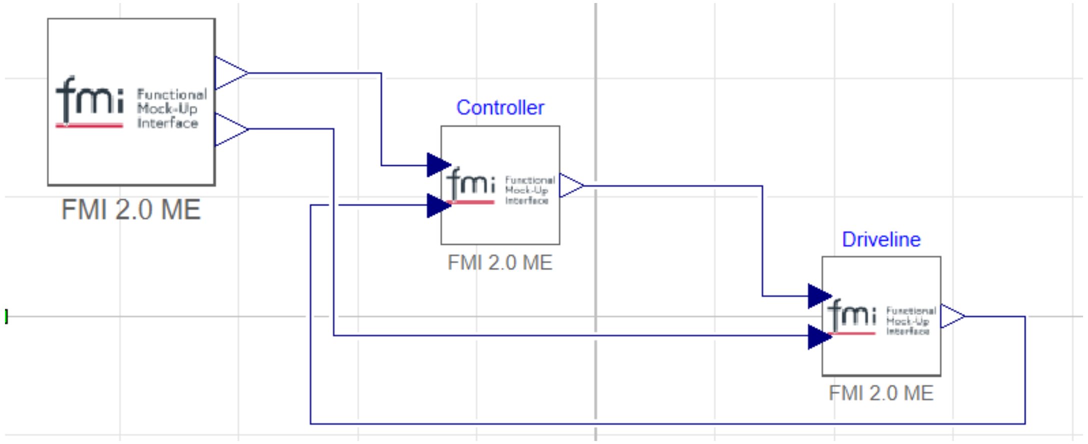
Demo: FMU Export and import

FMU Export from Modelica-tool Dymola:

- Configure options



FMU Import and Simulation in Dymola:



FMU Import and Simulation in MATLAB Simulink

- FMI import by Mathworks
  FMU import Block available in Simulink/extras